

NonTraditional_Mobility_Aid_Biomechanics_ParaTransit_Burden

June 5, 2026

1 Filed Exhibit A Reproducibility Notebook

This notebook is frozen as the reproducible source for the filed Exhibit A PDF:

Exhibit_A_Wearable_Biomechanical_ParaTransit.pdf

Original filed-generation command:

```
jupyter nbconvert --to pdf legal/notebooks/NonTraditional_Mobility_Aid_Biomechanics_ParaTransit
```

Do not use this notebook for ongoing research changes. Do not update its analysis logic, cohort definitions, tables, figures, or legal text except for intentional filed-record corrections.

For continuing wearable biomechanics, ParaTransit burden, and Supreme Court-level record development, use:

legal/notebooks/Wearable_Biomechanical_ParaTransit_Reproducible.ipynb

1.1 Evidence data loading and SHA-256 verification

This cell is designed to work in **PyCharm locally** and later in **Google Colab**.

It tries, in order:

1. A local file in the current notebook folder or repo.
2. Any file matching `fsicss_legal_corpus_categorized*.jsonl`.
3. A Colab upload path like `/content/fsicss_legal_corpus_categorized(1).jsonl`.
4. A raw GitHub URL, when `ALLOW_GITHUB_DOWNLOAD = True`.

The expected SHA-256 below is for the uploaded evidence corpus used in this repair:

`fsicss_legal_corpus_categorized.jsonl`

size bytes: 4505465

SHA-256: c7367238ffdc6e708343b297ffd82e1b305d4c312e5dccdfd30cbc145862527c

1.2 1. Input discovery

The notebook searches common Colab, local repo, and `/mnt/data` paths. If the corpus is missing in Colab, upload it when prompted.

1.3 2. Flatten records and validate schema

The notebook preserves nested source objects but also flattens common metrics into columns for reproducible calculations.

1.4 3. Cohort definitions

The comparator cohorts are deterministic and intentionally narrow.

- Walking is compared to Mall/PT controlled skating.
- FNS/SNS endurance skating is excluded from the primary walking comparator.
- ParaTransit bus/cutaway and van are compared to sedan.

1.5 4. WHOOP physiological context

WHOOP provides longitudinal heart-rate, HRV, recovery, and strain context. It is not used as the precise bus/van/sedan vehicle-type comparator.

1.6 5. Strava functional distance context

Strava shows functional distance and route context for walking and skating.

1.7 6. Kubios RRI/HRV+ and accelerometer comparator analysis

The Kubios accelerometer values below are **proxy movement-burden metrics**. They are relative comparisons using the same processing pipeline.

1.8 7. Charts

Kubios chart labels use proxy language to avoid overclaiming exact anatomical force.

1.9 8. Exhibit value reconciliation

This table compares calculated values against the draft Exhibit A reference values. Differences are flagged for review rather than hidden.

1.10 9. Legal language snippets

These snippets preserve the caveat while keeping the legal framing useful and honest.

1.11 10. README and zip outputs

This creates a small reproducible package with charts, tables, warnings, and legal language.